Week-2

Introduction to Database Systems

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How is a house constructed?

- If you want to build a house what steps will you follow?
  - We will give our requirements to the architect.
  - He will show us some paperwork i.e. a map or a 3d model.
  - Then the foundations are laid and grey structure is built according to the preferred material.
  - We then finalize about the finishing and furnishing.

Design Phases of Database

1. Requirement Definition and analysis

- Design of a new application for an existing database

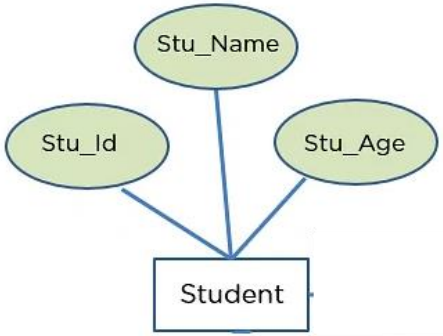
  OR

- Design of a new database

2. Conceptual Design

- These requirements are documented in detail to form a conceptual design
- Conceptual design can be represented and manipulated using some computerized tool so that it can easily be maintained, modified and transformed to database implementation.

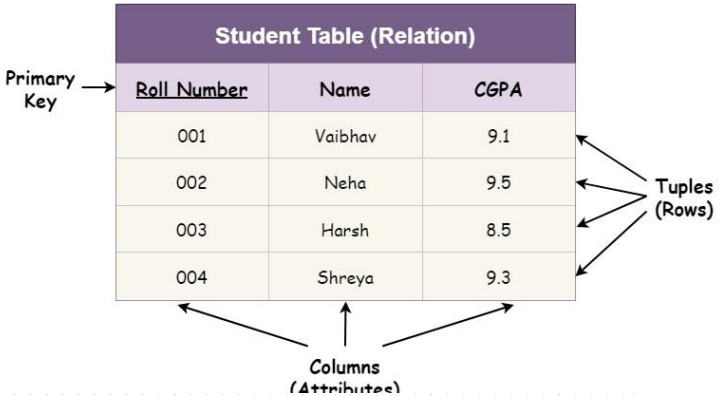
  Example: Entity Relationship Model



Design Phases of Database

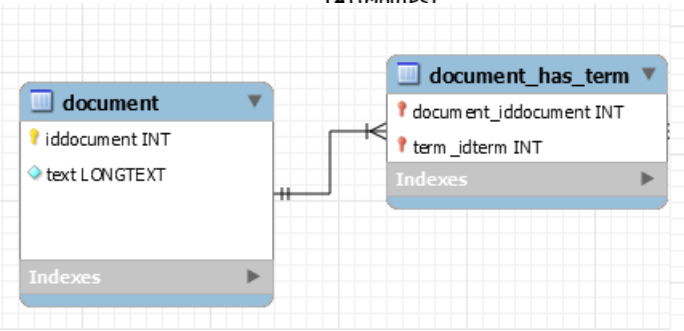
3. Logical Design

- Conceptual design is transformed into logical design
- Logical design can be expressed in a **data model**
- Example: Relational Data Model



4. Physical Design

- In this stage further specifications are provided for storing and accessing the database.
- Database design is implemented , populated with actual data and continuously maintained to reflect the actual state of miniworld.



Entity

Entity in Database

- An entity is an object that exists. It doesn't have to do anything; it just has to exist. In database administration, an entity can be a single thing, person, place, or object. Data can be stored about such entities.
- **Strong Entity**

  An entity is strong if it can independently exist

  Example: Student is a strong entity type as it has a unique ID for each student.
- **Weak Entity**

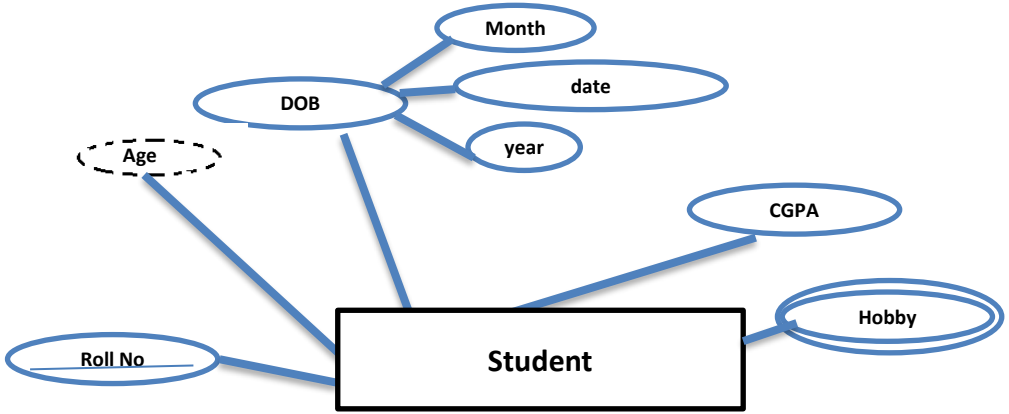
  An entity is weak when its existence depends on another entity.

  Example: Section can't exist without class

Attributes of an Entity

- An attribute is a property or characteristic of an entity.
- An entity may contain any number of attributes.
- One of the attributes is considered as the primary key.

Example:
a customer can have attributes such as name, email, and address.



Entity Set

An entity set is a collection or set of all entities of a particular entity type at any point in time. The type of all the entities should be the same.

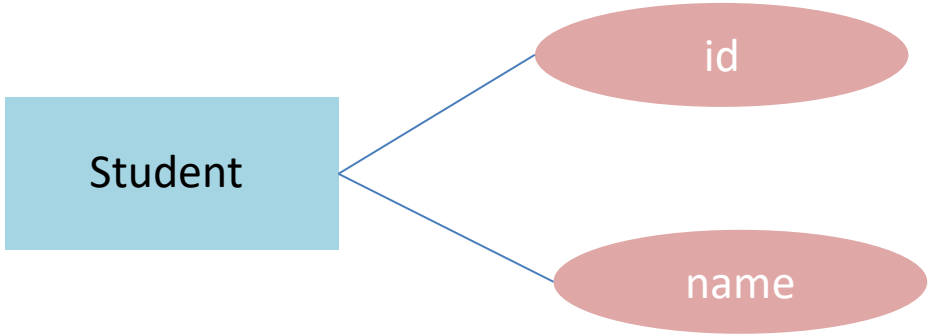
Entity Set

Example :

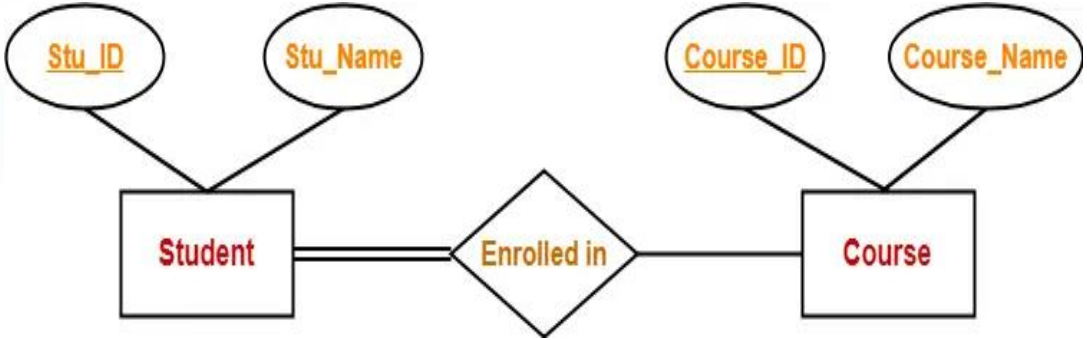
- The collection of all the students from the student table at a particular instant of time is an example of an entity set.
- The collection of all the employees from the employee table at a particular instant of time is an example of an entity set.

Entity Relation Diagram

- In an Entity-Relation model, attributes are represented in an elliptical shape.
- Properties of an entity



Entity Relation Diagram



Symbols Used for Strong Entity:

•A single rectangle is used for representing a strong entity set.

•A diamond symbol is used for representing the relationship that exists between two strong entity sets.

In this ER diagram:

•Two strong entity sets "**Student**" and "**Course**" are related to each other.

•Student ID and Student name are the attributes of entity set "Student".

•Student ID is the primary key using which any student can be identified uniquely.

•Course ID and Course name are the attributes of entity set "Course".

•Course ID is the primary key using which any course can be identified uniquely.

•Double line between Student and relationship set signifies total participation.

•It suggests that each student must be enrolled in at least one course.

•Single line between Course and relationship set signifies partial participation.

•It suggests that there might exist some courses for which no enrollments are made.

Entity Relation Diagram

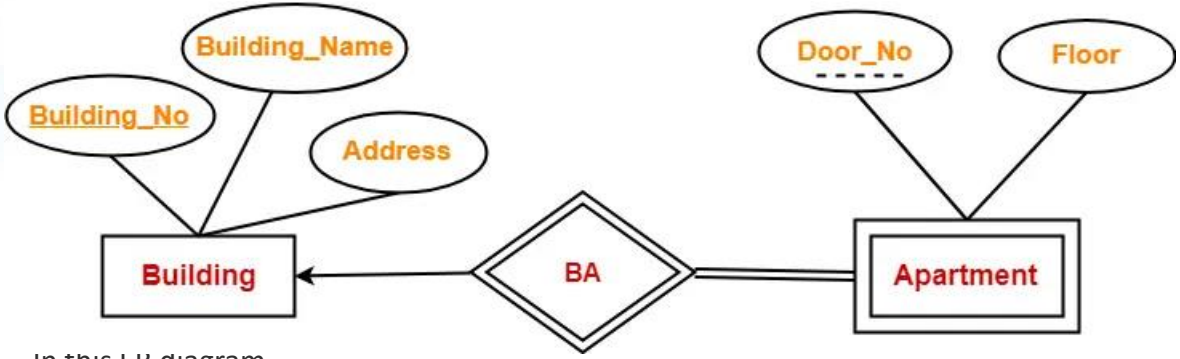
Symbols Used for Weak Entity:

•A double rectangle is used for representing a weak entity set.

•A double diamond symbol is used for representing the relationship that exists between the strong and weak entity sets and this relationship is known as **identifying relationship**.

•A double line is used for representing the connection of the weak entity set with the relationship set.

•Total participation always exists in the identifying relationship.



In this ER diagram,

•One strong entity set "**Building**" and one weak entity set "**Apartment**" are related to each other.

•Strong entity set "Building" has building number as its primary key.

•Door number is the discriminator of the weak entity set "Apartment".

•This is because door number alone can not identify an apartment uniquely as there may be several other buildings having the same door number.

•Double line between Apartment and relationship set signifies total participation.

•It suggests that each apartment must be present in at least one building.

•Single line between Building and relationship set signifies partial participation.

•It suggests that there might exist some buildings which has no apartment.

Types of an attribute

Types of attributes

- Key attribute [Student ID, CNIC, Passport Number]
- Composite attribute [The address can be further split into house number, street number, city, state etc]
- Derived attribute [Age can be derived from DOB]
- Simple attribute [Employee ID]
- Multivalued attribute [Contact number]

Keys and its Types

- A key in DBMS is an attribute or a set of attributes that help to uniquely identify a tuple (or row) in a relation (or table).
- Keys are also used to establish relationships between the different tables and columns of a relational database.
- Individual values in a key are called key values.

Keys and its Types

1. Primary Key
2. Super Key
3. Candidate Key
4. Alternate Key
5. Foreign Key
6. Composite Key
7. Unique Key

Keys and its Types

Primary Key

A primary key, also called a primary keyword, is a column in a relational database table that's distinctive for each record.

It's a unique identifier, such as a :

- driver's license number,
- telephone number with area code or
- CNIC.

Keys and its Types

Super Key

A super key is a superset of a candidate key.

For example: In EMPLOYEE table, for(EMPLOEE_ID, EMPLOYEE_NAME), the name of two employees can be the same, but their EMPLYEE_ID can't be the same.

Hence, this combination can also be a key.

The super key would be EMPLOYEE-ID (EMPLOYEE_ID, EMPLOYEE-NAME), etc.

Keys and its Types

Candidate Key

For example, employee relation has two attributes, Employee_Id and NTN, that act as candidate keys.

In this relation, Employee_Id is chosen as the primary key, so the other candidate key, NTN, acts as the Alternate key.

Keys and its Types

Alternate Key

Let's take an example of student. It may contain NAME, ROLL NO., ID and CLASS. Here ROLL NO. is primary key and rest of all columns like NAME, ID and CLASS are alternate keys.

If a table has more than one candidate key, one of them will become the primary key and rest of all are called alternate keys.

An alternate key is a column or group of columns that uniquely identify every row in a table.

Keys and its Types

Foreign Key

A foreign key is a column or group of columns in a relational database table that provides a link between data in two tables. It is a column (or columns) that references a column (most often the primary key) of another table.

Example: STUD_ID in STUDENT_COURSE is a foreign key to STUD_ID in STUDENT_INFO table.

Keys and its Types

Composite Key

A composite key, then, would be the combination of two keys.

For example: the combination of house number and street might qualify as a composite key, given that the market listings are local.

If so, then when someone searches using both the house number and the street, they should only get one single record returned.

Keys and its Types

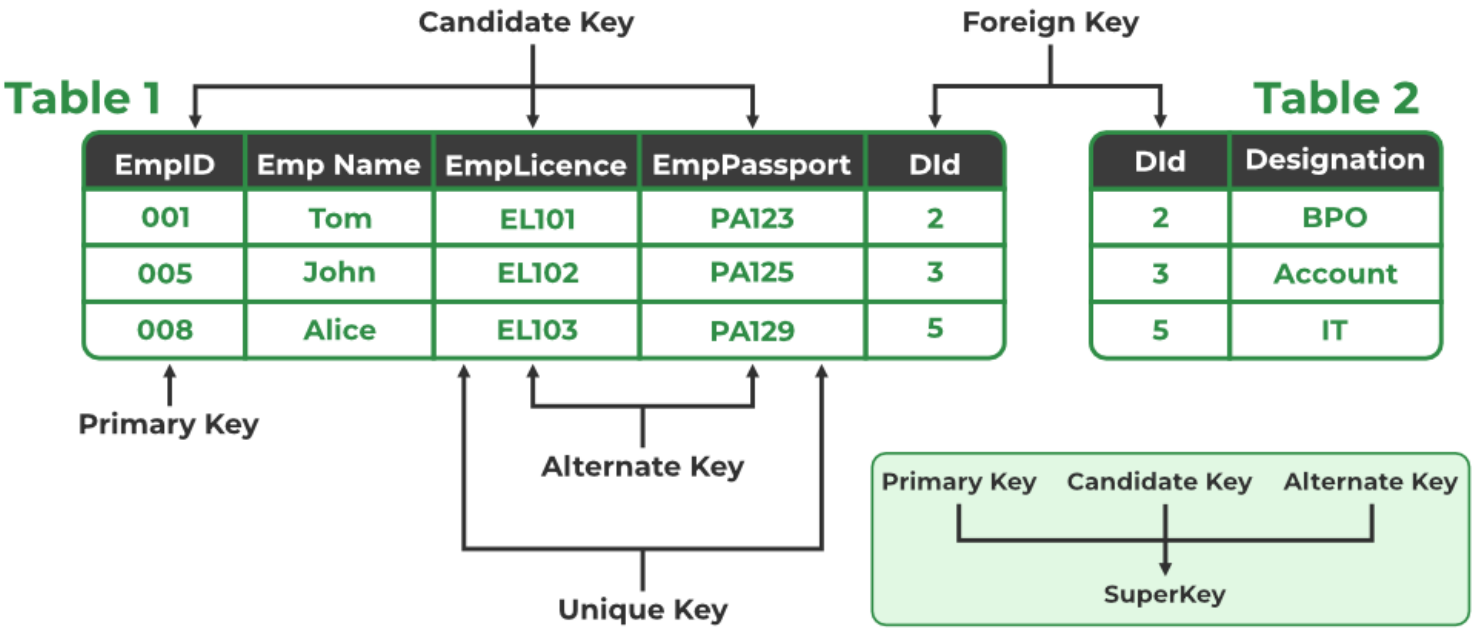
Unique Key

Let's see an example using the Employees table.

Each employee must have a unique number; hence, the EmployeeNumber column implements the UNIQUE constraint.

By doing so, we ensure that no employees in the same company have the same employee number.

Keys and its Types

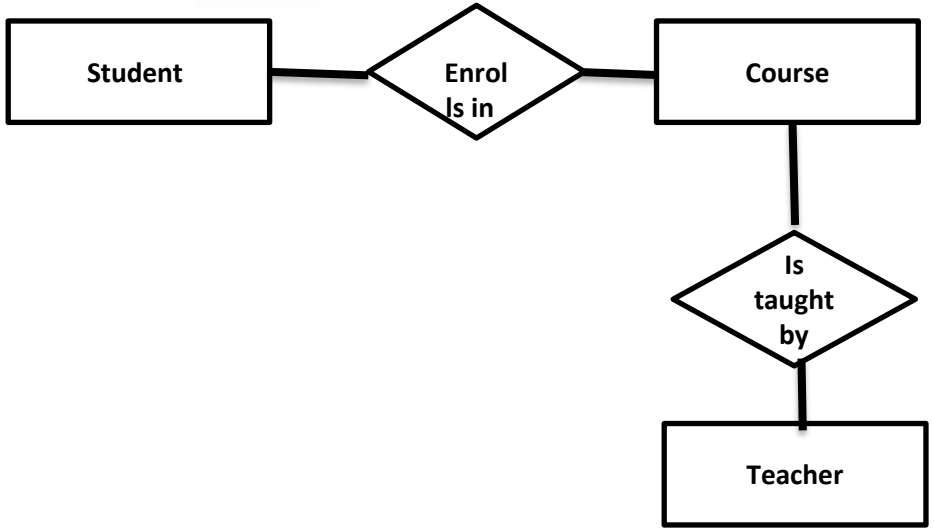


EmpID	Emp Name	EmpLicence	EmpPassport	DId
001	Tom	EL101	PA123	2
005	John	EL102	PA125	3
008	Alice	EL103	PA129	5

DId	Designation
2	BPO
3	Account
5	IT

Relationship representation in ERD

Please note: Readability is Left to Right & Top to bottom



Constraints on Relationship

Cardinality

- One to many (1:N)
- One to one (1:1)
- Many to many (M:N)

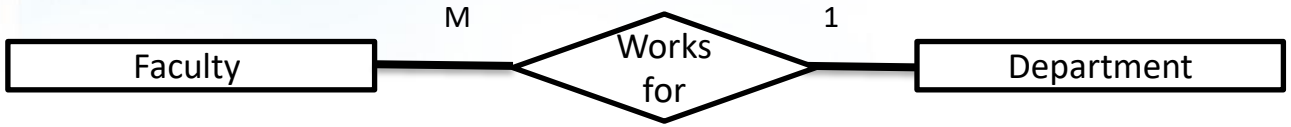
Participation

- Total Participation
- Partial participation

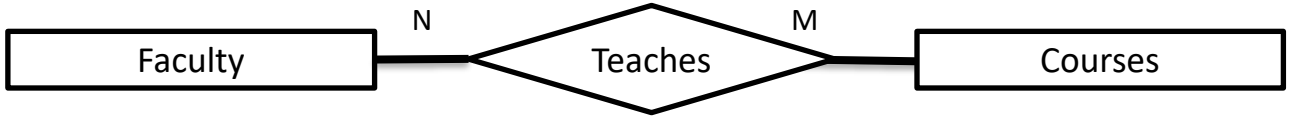
Relationship: cardinality

Cardinality

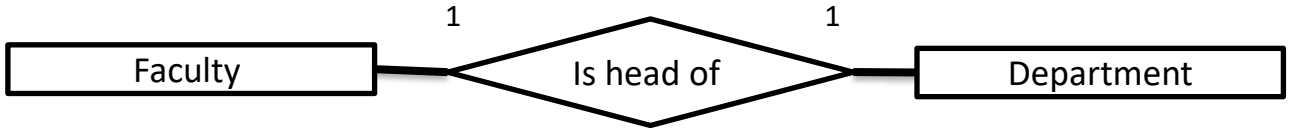
- 1 : M



- M : N

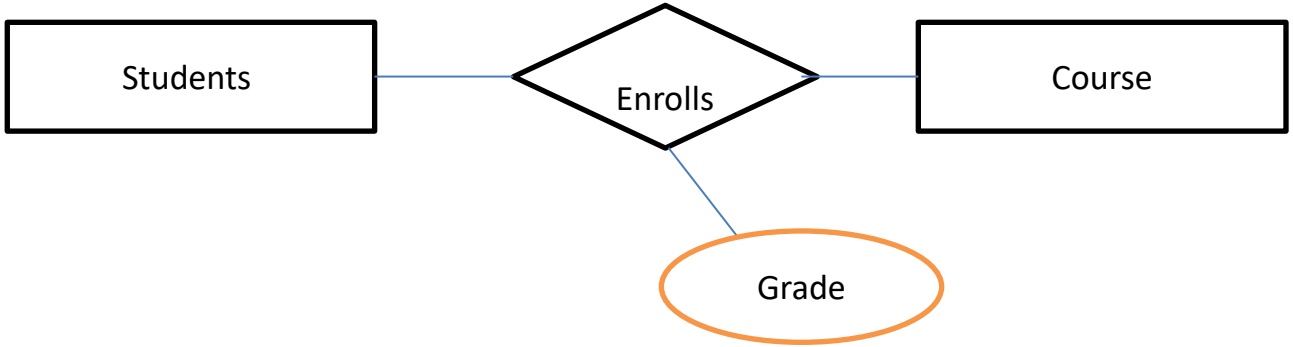


- 1 : 1



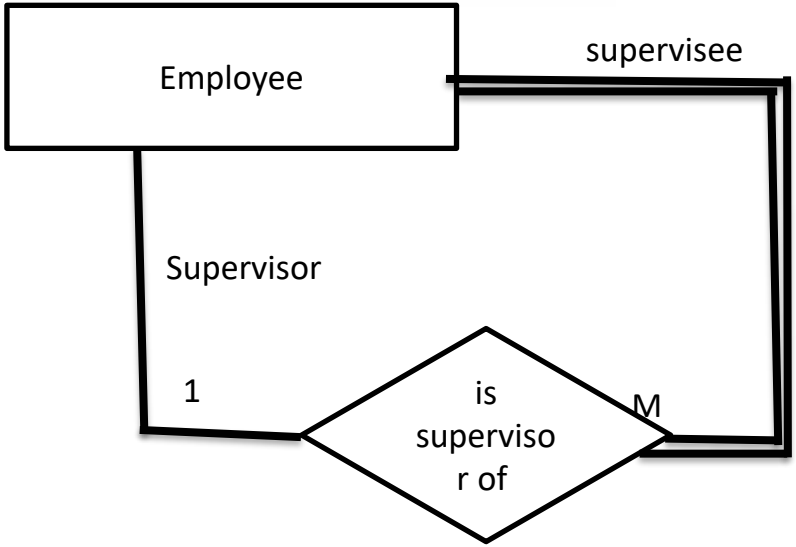
Attributes of relationship

- A relationship can have attributes:
  - For example, grade of enrolls
  - Its value for each relationship instance describes the grade a student takes when he is enrolled in a course.
    - A value of Grade depends on a particular (Student, Course) combination



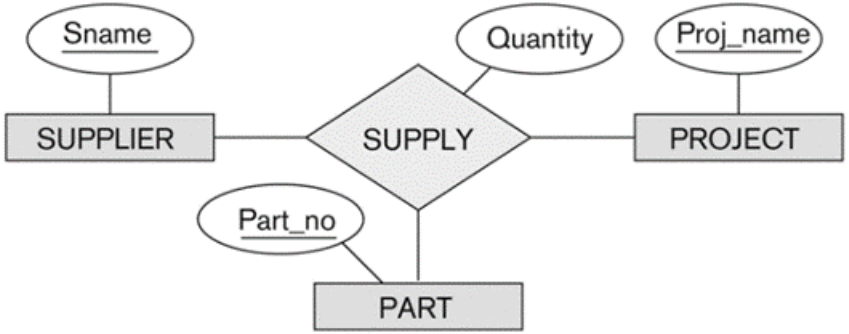
Recursive Relationship: Rolenames

- Employee is supervisor of other employees



Ternary relationship

(a)



Activity: Design an ERD

- The company is organized into departments. Each department has a unique name, a unique number, and a particular employee who manages the department. We keep track of the start date when that employee began managing the department. A department may have several locations.
- A department controls a number of projects, each of which has a unique name, a unique number, and a single location.
- We store each employee's name, Social Security number,2 address, salary, gender , and birth date. An employee is assigned to one department, but may work on several projects, which are not necessarily controlled by the same department. We keep track of the current number of hours per week that an employee works on each project. We also keep track of the direct supervisor of each employee (who is another employee).
- We want to keep track of the dependents of each employee for insurance purposes. We keep each dependent's first name, gender, birth date, and relationship to the employee.

Mention cardinality & participation
Identify Weak entities
Identify relationship attributes

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Thank You all!